

Improving Korean NLP Tasks with Linguistically Informed Subword Tokenization and Sub-character Decomposition

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- [paper link](#)

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Abstract

We investigate whether Transformer-based language models, trained solely on Hangul text, can learn the compositional morphology of Sino-Korean (SK) morphemes. Using BERT_{BASE} and fastText, we conduct controlled experiments with target words and their “real” vs. “fake” neighbors—pairs that share a Hangul syllable representing the same SK morpheme vs. those that share only the Hangul syllable. Our results show that, while both models—especially BERT—distinguish real and fake pairs to some extent, performance is largely driven by the frequency of each experimental word rather than a true understanding of SK morphemes. These findings highlight limits of distributional learning for morpheme-level understanding and motivate explicit morphological modeling or Hanja-aware strategies for Korean. Dataset and code: <https://github.com/taeheejeon22/ko-skmorph-lm>.

Dataset Overview

A compact corpus of Sino-Korean (SK) two-syllable words curated to test whether language models can recover morphemic meaning from Hangul alone. For each of 100 targets, we provide two neighbors: a **real** neighbor that shares the same SK morpheme in the first syllable and a **fake** neighbor that shares only the Hangul syllable (no SK morpheme match). The analysis-ready CSV augments these pairs with similarity scores (BERT/fastText), Top-K similar-word counts/ratios (K=500/600), and frequency features used in

Experiments 1–3.

	Hangul	Hanja	meaning
Target Word	신작	新作	<i>new work</i>
Real Neighbor	신입	新入	<i>newcomer</i>
Fake Neighbor	신용	信用	<i>trust</i>

Source Dataset

- `data/dataset.xlsx` — Source dataset

Schema

File: `data/dataset.xlsx` (Excel, Sheet: `Sheet1`)
Rows: 100 targets (each with a real and a fake neighbor)

Column	Type	Domain / Range	Description
<code>target_word</code>	str	Hangul	Target word (표제어).
<code>target_hanja</code>	str	Hanja	Hanja form of the target.
<code>morpheme_structure</code>	str	{ <code>Bound</code> + <code>Bound</code> , <code>Bound</code> + <code>Free</code> }	SK morpheme structure of the target .
<code>target_headword</code>	str	Hangul	Target lemma/headword (often same as <code>target_word</code>).
<code>PoS</code>	str	e.g., <code>Noun</code>	Part of speech for the target.
<code>num_homographs</code>	int	0–1	Count of homographs for the target .
<code>real_neighbor</code>	str	Hangul	“Real” neighbor sharing the same SK morpheme in the first syllable.
<code>real_hanja</code>	str	Hanja	Hanja form of the real neighbor.

Column	Type	Domain / Range	Description
<code>real_neighbor_num_homographs</code>	int	0–1	Homograph count for the real neighbor.
<code>fake_neighbor</code>	str	Hangul	“Fake” neighbor sharing only the Hangul syllable (no SK morpheme match).
<code>fake_hanja</code>	str / null	Hanja or empty	Hanja form of the fake neighbor (often empty when not SK).
<code>fake_neighbor_num_homographs</code>	int	0–1	Homograph count for the fake neighbor.
<code>example_target</code>	str	serialized list	Example sentences for the target (stringified list of strings).
<code>example_real</code>	str	serialized list	Example sentences for the real neighbor.
<code>example_fake</code>	str	serialized list	Example sentences for the fake neighbor.

Notes

- `fake_hanja` is missing for some rows where the fake neighbor is **not** an SK word (21 blanks observed).
- Example fields are stored as **stringified lists** (e.g., `"[['...', '...']]"`)
- This Excel file is the **source** for the analysis-ready CSV; no model-derived metrics are included here.

Analysis-ready File

- `data/dataset_for_experiments.csv` — Analysis-ready CSV used in all experiments. Encoding: UTF-8 (utf-8-sig).

Schema

Core identifiers

Column	Type	Domain / Range	Description
<code>index</code>	int	0–199	Row index (provided in file).
<code>real_fake</code>	int	{1, 0}	1 = Real pair (shared SK morpheme), 0 = fake pair.
<code>morpheme_structure</code>	str	{ <code>Bound + Bound</code> , <code>Bound + Free</code> }	SK structure category of the target .

Column	Type	Domain / Range	Description
PoS	str	e.g., 명	Part of speech for the target .
num_homographs	int	0–1	Count of target homographs.

Target / neighbor descriptors

Column	Type	Domain / Range	Description
target_word	str	—	Target word (Hangul).
target_hanja	str	—	Target’s Hanja form.
target_headword	str	—	Lemma/headword of target.
neighbor_word	str	—	Neighbor word paired with target.
neighbor_hanja	str	—	Hanja form of the neighbor.
neighbor_num_homographs	int	≥0	Homograph count for neighbor.

Similarity & decision variables

Column	Type	Domain / Range	Description
cos_sim_neighbor_BERT	float	[-1, 1]	Cosine similarity (BERT) between target and designated neighbor .
cos_sim_neighbor_ft	float	[-1, 1]	Cosine similarity (fastText) for the same pair.
cos_sim_real-fake_BERT	float	[-1, 1]	Cosine difference used in Exp. 1 (BERT).
cos_sim_real-fake_ft	float	[-1, 1]	Cosine difference used in Exp. 1 (fastText).
correctness_BERT	int	{0,1}	1: BERT chose the real neighbor.
correctness_ft	int	{0,1}	1: fastText chose the real neighbor.

Frequency features (raw / relative)

Column	Type	Domain / Range	Description
freq_BERT, freq_ft	int	≥0	Token frequency of target in each model’s corpus proxy.
freq_BERT_relative, freq_ft_relative	float	≥0	Relative frequency of target (normalized).

Column	Type	Domain / Range	Description
<code>neighbor_freq_BERT, neighbor_freq_ft</code>	int	≥ 0	Token frequency of neighbor.
<code>neighbor_relative_freq_BERT, neighbor_relative_freq_ft</code>	float	≥ 0	Relative frequency of neighbor (normalized).
<code>real_neighbor_freq_token_BERT, real_neighbor_freq_token_ft</code>	int	≥ 0	Token frequency of real neighbor.
<code>freq_real_neighbor_token_BERT_relative, freq_real_neighbor_token_ft_relative</code>	float	≥ 0	Relative frequency of real neighbor.

Similar-words retrieval (Top-K = 500/600)

Replace `*` with `BERT` or `ft`; replace `{K}` with `500` or `600`.

Pattern	Type	Description
<code>num_sim_words_neighbor*_n{K}</code>	int	Count of Top-N similar words that share only Hangul with the target.
<code>num_true_sim_words_neighbor*_n{K}</code>	int	Count of Top-N similar words that share both Hangul and Hanja .
<code>num_true_sim_words_neighbor*_n{K}_hanja</code>	int	Same as above, explicitly filtered by Hanja match.
<code>ratio_true_sim_words_neighbor*_n{K}</code>	float	Proportion of true similar words among Top-N (0–1).
<code>ratio_true_sim_words_neighbor*_n{K}_hanja</code>	float	Proportion (0–1) when restricted to Hanja match.

Examples & embeddings (for transparency; not needed to run stats)

Column	Type	Description
<code>example_target, example_neighbor</code>	str	Example sentences for target / neighbor.
<code>example_target_tokenized_BERT, example_neighbor_tokenized_BERT</code>	str	Tokenized examples (BERT).
<code>vector_target_BERT, vector_target_ft</code>	str	Serialized embedding vector for target (diagnostic).
<code>vector_neighbor_BERT, vector_neighbor_ft</code>	str	Serialized embedding vector for neighbor (diagnostic).

Column	Type	Description
num_examples_target, num_examples_neighbor	int	Number of example sentences used.

Environment

- Python 3.9–3.11
- See requirements.txt
- Tested on macOS 15.6.1

Reproducing the Experiments

1. (Recommended) Create a virtual environment and install dependencies
2. Run scripts/experiments.ipynb from the repository root. For example, in VS Code, set **Jupyter: Notebook File Root** to \${workspaceFolder} in **Settings**, then restart the kernel.

Citation

- TBA

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